NEW BEST-KNOWN MAX-k-CUT SOLUTIONS ON THE G-SET BENCHMARK

A PREPRINT

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ABSTRACT

We report seven new best-known Max-k-Cut values on the G-set benchmark, relative to the published results identified as of October 15, 2026. For G81 with $k = 3$, we obtain 16,510, exceeding 16,392 by 118. For G22 with $k = 4$ and G55 with $k = 3$, we obtain 18,787 and 12,437, improving the published values by 11 and 5, respectively. For $k = 5$ on G1–G4, we obtain 17,706, 17,710, 17,705 and 17,710, improving the published values by 1, 3, 2 and 1. The accompanying release contains 35 explicit vertex partitions: these seven improvements, 17 matches to published best-known values (including three provably optimal solutions), ten reference values for $k = 6, 8$, and one additional G81 partition for $k = 4$. The latter has value 16,666, below the published Gurobi value of 16,670, and is not claimed as a new best-known result. This note publishes solution certificates and their verification, while leaving the solution-generation algorithms to a separate study.

Keywords Max-k-Cut · G-set · graph partitioning · solution certificates.

1 Introduction

The G-set graphs are a widely used benchmark for Max-Cut and its generalisation, Max-k-Cut [1, 2]. Given a weighted undirected graph $G = (V, E, w)$ and $k \geq 2$, the objective is to assign each vertex to a part so as to maximise

$$C(x) = \sum_{(i,j)\in E} w_{ij}[x_i \neq x_j], \qquad x \in \{0, \ldots, k-1\}^{|V|}. \tag{1}$$

All certificates in this release use all k labels. A feasible partition gives a lower bound on the optimum and can be checked directly from the edge list, independently of how it was found.

This note documents seven improved values and their partitions. Its contribution is the set of verifiable solutions and the comparison with published values; it makes no comparative runtime or algorithmic performance claim.

2 Published comparison values

The multiple search operator heuristic (MOH) of Ma and Hao [2] reports G-set values for $k = 2, 3, 4, 5$. The $k = 5$ comparisons here use the accepted journal manuscript rather than the duplicated $k = 4$ table in arXiv version 1. Garvardt et al. [3] evaluated parameterized local search (LS) and integer linear programming (ILP) on MOH starting partitions. Their 2026 journal version reports 16,392 for G81 with $k = 3$ (LS, Table 7), 16,525 for G81 with $k = 4$ (LS, Table 8), and 12,432 for G55 with $k = 3$ (ILP, Table 7).

The parallel evolutionary algorithm framework (PEAF) of Shao and Yang [4], published online in September 2026, reports the values used here for G1–G4 with $k = 5$ (Table 21), as well as several of the matched values (Tables 19–20). The population annealing study of Khan et al. [5] also reports 12,432 for G55 and 17,169 for G22 with $k = 3$ (Table S2).

It is necessary to compare against all reported solver columns. In particular, Shao and Wu [6] report a Gurobi value of **16,670** for G81 with $k = 4$ in Table 3, higher than the LS value of 16,525. This published value is included in our comparisons. No explicit partition for that external value was independently checked in this study.

3 Results

3.1 Seven improved best-known values

Table 1 lists the improvements over the highest published values identified in the literature reviewed for this note. None of these seven solutions is claimed to be globally optimal.

Table 1: Improved Max-k-Cut values. Δ is the gain over the published comparison value.

Graph	k	This work	Previous	Δ	Gain (%)	Source
G81	3	**16,510**	16,392	+118	0.720	LS [3]
G22	4	**18,787**	18,776	+11	0.059	MOH, PEAF [2, 4]
G55	3	**12,437**	12,432	+5	0.040	ILP, PAMC [3, 5]
G1	5	**17,706**	17,705	+1	0.006	PEAF [4]
G2	5	**17,710**	17,707	+3	0.017	PEAF [4]
G3	5	**17,705**	17,703	+2	0.011	PEAF [4]
G4	5	**17,710**	17,709	+1	0.006	MOH, PEAF [2, 4]

G81 has 20,000 vertices and 40,000 edges, including 19,983 edges of weight -1. Its objective is the signed sum in Eq. (1), not the number of crossing edges. G22 has 2,000 vertices and 19,990 edges; G55 has 5,000 vertices and 12,498 edges; G1–G4 each have 800 vertices and 19,176 edges. The latter six graphs have only unit positive weights.

3.2 Seventeen matches and three optimality certificates

Table 2 gives 17 matches to published values. The three optimal cases are included in this count. In each of these cases all edge weights are $+1$ and every edge crosses the partition, attaining the upper bound $|E|$.

Table 2: Partitions matching published best-known values. Values on G1–G5 are in graph order.

Graph	k	Cut values	Comparison
G1–G5	3	15,165, 15,172, 15,173, 15,184, 15,193	MOH, PEAF [2, 4]
G1–G5	4	16,804, 16,810, 16,806, 16,814, 16,816	PEAF [4]
G5	5	17,711	PEAF [4]
G22	3	17,169	PEAF, PAMC [4, 5]
G43	3, 4	8,573, 9,377	MOH; LS/ILP, PEAF [2, 3, 4]
G55	4	12,498	All edges cut; optimal
G70	3, 4	9,999, 9,999	All edges cut; optimal

3.3 Ten reference values

We did not identify directly comparable published values for G1–G5 with $k = 6$ or $k = 8$ in the literature reviewed. Table 3 therefore provides reference points, without making a claim of priority.

Table 3: Reference values on G1–G5 without an identified published comparison.

k	G1	G2	G3	G4	G5
6	18,255	18,257	18,254	18,257	18,258
8	18,843	18,844	18,843	18,844	18,845

3.4 Additional G81 certificate

The release also contains our G81 partition for $k = 4$, with value 16,666. It is four below the published Gurobi value of 16,670 [6] and is listed separately from both improvements and matches. Including this partition brings the total number of certificates to 35.

4 Certificate encoding and verification

Every partition is provided as a UTF-8 text file with one integer per line: line i is the label of vertex i, in the original G-set numbering. The file `results.csv` records graph size, k, cut value, comparison source, and the SHA-256 digest of each certificate. Original graph files are obtained from the Stanford G-set directory [1]; their expected hashes are included in the release.

The dependency-free Python verifier checks graph and solution hashes, the number and range of labels, and Eq. (1). An independently written checker additionally confirms that no single-vertex move improves any of the 35 partitions. This local optimality does not establish global optimality. The three all-edges-cut cases have the separate upper-bound proof given above. The checks verify the solutions, not the completeness of a literature search.

Appendix A embeds the seven improved partitions so that these certificates remain available with the paper. For $k \leq 4$, expand each hexadecimal digit to four bits, most significant bit first, and read each successive pair of bits as one vertex label. For $k = 5$, each hexadecimal digit is the label of one vertex. Concatenate wrapped lines before decoding.

5 Conclusion

The seven improved values comprise G81 with $k = 3$, G22 with $k = 4$, G55 with $k = 3$, and G1–G4 with $k = 5$. The explicit partitions establish their values without access to the solution-generation algorithms. The accompanying data also provide 17 matches, ten reference values and one additional G81 certificate, with the comparison categories kept distinct.

Data availability

The 35 partitions, comparison table and verification programs accompany this preprint as dataset version 2.0.0, with DOI `10.5281/zenodo.23405852`.

Repository: `https://github.com/yaochengchen/gset-maxkcut_by_PdSB`.

The DOI is reserved for this release and becomes active when the Zenodo record is published.

Acknowledgments

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Conflict of interest

The author declares no conflicts of interest.

References

[1] C. Helmberg and F. Rendl. G-set: Max-Cut benchmark instances. Generated with RUDY by G. Rinaldi; hosted by Y. Ye. `https://web.stanford.edu/~yyye/yyye/Gset/`. Accessed October 15, 2026.

[2] F. Ma and J.-K. Hao. A multiple search operator heuristic for the max-k-cut problem. *Annals of Operations Research*, 248:365–403, 2017. doi:10.1007/s10479-016-2234-0. arXiv:1510.09156; numerical comparisons use the accepted journal manuscript.

[3] J. Garvardt, N. Grüttemeier, C. Komusiewicz and N. Morawietz. Parameterized Local Search for Max c-Cut. *Annals of Operations Research*, 2026. doi:10.1007/s10479-026-07426-0. Published September 17, 2026; preliminary version in IJCAI 2023, pp. 5586–5594.

[4] S. Shao and C. Yang. A Parallel Evolutionary Algorithm Framework for Graph k-CUT Problems. *Journal of Intelligent Algorithms and Scientific Computing*, online first, 2026. doi:10.4208/jiasc.2026-194-1. Published September 28, 2026. arXiv:2607.17158.

[5] N. Khan, R. Redding and N. Shukla. Leveraging Population Dynamics to Steer Efficient Search in Large-Scale Combinatorial Optimization. arXiv:2606.25203v1, 2026. `https://arxiv.org/abs/2606.25203`.

[6] S. Shao and Y. Wu. An ODE Approach to Multiple Choice Polynomial Programming. *East Asian Journal on Applied Mathematics*, 15(1):1–28, 2025. doi:10.4208/eajam.2023-184.151023. arXiv:2212.06371v2.

A The seven improved partitions

G81, $k = 3$, cut 16,510

SHA-256 of the plain-text certificate:


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G22, $k = 4$, cut 18,787

SHA-256 of the plain-text certificate:

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 ae43a55f5c91cb77ce431d863fe21829b307de3e7dba9c2237621d69f38de4c4f166af9beeceb4d56811aee441bc8e9c7c8a
c580ef60589d92f60db9dcfecb4a82475ce4ba696024f76291a70ed34031b66034ccfaf1942b713b17c9de5220ea21df0d05
6879e8b36f3b778cbad985ed9f5dc44f463cdd28f334b0b557c340e1508ff2674f0f08e53c35ae21c28ba5f6acf4b9829a5d
4692589edd05c3a18005e86571bf00a6aa5842d62f8169f009cf2234ca7eacd98db829b973246102dd5fdbfd084c9c4b6f8f
1449afaf316a00505ebe7bde59d55762ed458493fbc96b36d40538b3aa2087f9404c3bd4e50d2999ac4b82ad49df86b05844
0f4d868f0cb3ac960ef66a191c6ed42ed79016497b47a51bcb89011553cd1b627d000c2143c03eab4f9e38e9fd2e12ca0882
3cdb20c1d8d63e54a4e97c72c98f46ff52996906e1d4c422c276d9e6bb0bb96972f5ec084e5644970f07335d221d4e7842e4
dd5acec036ca407bc49fedffa15832b0f576bd836774238f4eb6ea8f4d16e4e86af9ddc70e7491ee5d6b38a19e4ba6792f1f
d1bdbdb735b5b9dd38319e8994c107e29300bd89adb435fe4eb22b44927b8eaf21c4def42e18343dd3a035412234775e4e85
6dab468ce136e2a783a88d9ff3f4dfd8b5cf6540590837c95ea573ca12f6726839d1d8e1796abea7565585a3c40643a6ee7e
```

G55, $k = 3$, cut 12,437

SHA-256 of the plain-text certificate:

```
8281545f21ada626a8f3016f93698196649393a5a2083b327fa865641199f07b
 855959201419024611a589a2992490848488985158 29a610211196a64942a652a8818965819055866aa125054268200a5960
a1662088991a161656401a558141a901996a095811a2100822229aaa1090164250564 5a648956985044a4429255a90126a18
8a6001256595524860850a9555256902846062854292405821a560580a6195a0a2a5a0a625119920919a1a29499522156681
98099a9816606654886588 8a6221848528119125528 0a1969aa06009068044851a6116894941a4886469a6a4a66824620501
5500a94a5228491466169040264902214580406596 84a68288a2618a1428411009240849550180418624 4a52a8822554a16a
a198669a186465a026010688824699 80a110aa6850169240148886258120 0a6160941085566942065 6a142810222966a8899
946629258560841681952905680888286295942619588461658101404510284192 09a01152a28521004a5012a850946552998
80944a28a460460659981028a960204450026a5809a9858160a0a24a2246611aa292486645644856684666 4a29005816a411
594691aa9202128158a5a6818004459114118aaa018412a880692891459909405999524468000602416166502 5a02110aa91
98052698a5a9a894699010a490552481a188a2552528aa254025606a269688a98aa404496a0804280614569129289046491 1
924a8aaaa248499a04945aa96a6855564a425515884956014aa6455859aa9496506a20596044460098a044a88262aa542aa4
092a52019465a81aa9585121948984141a1024a0020024195486498941142691a594860064452486298411008404614511 00
a1a98621159580a8122aa4a09489a8254058840294aa00a4484010a168a9562182a6215686082068629816884544564 1655a
804900a4552146a1084164629909691 29aa1455892aa85a258502199a651451084605 0a6a2864aa59046019a559145898a26
66526a180090014891a4429a620856964586aa2945415a104611846a24a988981619819a228102925684099495059404502 2
6215495441604080445008599026695 84a2906254920025986a8691845461450820064a85a29415868028a96410448505459
a5a4104a1021992050416554592922825682486949429 5a226682a009119609692a692908096a2995089546a518a81490454
4a92412988858884621211121291105940050521a144162a0059a6a4651054040645644282912a989690208aa8841466a168
65a0202415595050408a88694500209625085 19a6695825122822808849888a8844228942a4640919189a1140154a908662a6
859451161a6022851a1182601559952 61a81848682a82a5598421a1902559229a289862581a691205864a810260246a45844
909819a8a6882442156214999206a682281121441a248649696216101a96a442646860a65264095a965a09984859444a8924
608a40159881a86950008a5048a061969990696a269565954540605a0586025464691144509a121258695891452061598924
a8405008411228 98a460510850a9649652425065a909814254425416895a186616a8199011109a95a0648889a49a0209552a
88114a1650a6166468925a9592868294691004092569484 1a60062589661952a498a0224468556a19188528121220a050650
1129010955aa8a158a161a426585809522aa5450a8846446222890910082152006681052241058999046942560162085006a
```

G1, $k = 5$, cut 17,706

SHA-256 of the plain-text certificate:

```
13ca9b220557f8ad6a9605aa8bf0153ff38f1bb4a0c639f7b095c59a60166dc5
 331300311412430320222310031221201123322233443114300403031144321401314200444411014220044144244322243 2
204242403114124331222144023232001404334411140402324132410323333302234330321142334101313240404110004 1
012400233121433140001443310414324322001342113210342430303441300410011311411144303024122314401324344 4
423230034431323000321423000132302222102000323140331143024403034131230303140144204214214101120134122 2
343410210032014240423320301120413142411041230004413213123120022211203012340024313230220122042302200 1
131420302402132123444141241242214230012120021030204344223433321021443020430303212212010413432223103 0
221434032132441240002333223210014214310444340132412302232040113411401030330230223404210041143240414 2
331113342003123124124231340420442404411430240441404421443133131300400414310132104102220331303211002 1
```

G2, $k = 5$, cut 17,710

SHA-256 of the plain-text certificate:

```
f27c1a4c835c6588f7292efc5fee7b008607759f330c551c0faee6738a03393e
 131243032440300034422202114043203100002423043301020100331044420122002413012133040144441104233202120 0
014222140120034004144344323221320331322031044122044222414433434231124203440331311100421110301331110 2
113120112413303021321021103014203312101203401142010444003221141132100113400114443303224441431440341 3
012401131324200224003033412330213404024331234212013342042413102040421244014132221200401123424204344 0
122233103121332301034204114014414222310222441241322422133312132312140342023414433243220432042002111 1
334333104341213122034023023113200404230142133044344334120003421113200233430213123334114413301403440 4
431010324233242114101131044242034342000204443034214423320124344143421021001022111334114032234424023 2
402104124132314234300330431420120132131112031434004122241133320020130240103104341244220203014200421 4
```

G3, $k = 5$, cut 17,705

SHA-256 of the plain-text certificate:

```
a058b7591a5c3785059af04c719ab4e04340c3e01373a44340c2d198813675ca
 322122232030430041102130324101312020442340444242020102111010040224413403131240333304323332102040230 2
202023114414304023331141432101213101310442403424301112301412030130131341434021000044314040004344020 4
424003331130311104123303113034132141211044403223412200110314143130313411223343410402023102021424423 3
202202242442140114032044443321411012241430421341424043243302421130242104032443311130204314422123042 0
204323240130320422200321223442223401031031131413014311130033412141044431020141324121111221020223340 3
001224122413042332441431340434111024000442340444421032413333431210131402102230122212230312330001012 3
403332210330031130034333400220003222213033244412110131013031432101034100142041403141042034342220134 3
204241022442243132334411203230222132212301302330303234430400201201022243214202403012000034420141321 42
```

G4, $k = 5$, cut 17,710

SHA-256 of the plain-text certificate:

```
05edaa083eee7eca51cd6d5b23504d32366fffb186bd3208f79bfe5920cecbad
 110404423221334214303401144100102322443033204240013422242421201230201030334134023244022314430312014 0
324221431341132111334112232012443403230200242133400331313230022020040330201413310223022221134242403 2
323123121344204213211001313114034410003212440431314331323130303400120442102402104342411330201003300 4
234004110004203204242011144312423244241400222421043210200334422011141331043140131212011211242131200 2
242231301113300003104142310440342104443422344041323304013331004301010130423242132233441044244143311 4
022432441144002333434401044130204230010024103210344032333243331402302133110231012230213243003010222 0
131313342111412443404000223230441224440402304131434422114243000431043313421034133132111132320404243 4
233410102001212140403010042442140044304203311202110442401204111421423333240113204322123011234224301 1
```